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DO CONDITIONAL CASH TRANSFERS INFLUENCE MIGRATION? A STUDY USING EXPERIMENTAL DATA FROM THE MEXICAN PROGRESA PROGRAM*

GUY STECKLOV, PAUL WINTERS, MARCO STAMPINI, AND BENJAMIN DAVIS

Prior research on Mexican migration has shown that social networks and economic incentives play an important role in determining migration outcomes. We use experimental data from PROGRESA, Mexico's primary poverty-reduction program, to evaluate the effects of conditional cash transfers on migration both domestically and to the United States. Our study complements a growing body of literature aimed at overcoming longstanding hurdles to the establishment of causal validity in empirical studies of migration. Analysis based on the data collected before and after the program's onset shows that conditional transfers reduce U.S. migration but not domestic migration. The data also enable us to explore the role of existing family and community migration networks. The results show that migration networks strongly influence migration, but that the effect of conditional transfers on migration is apparently not mediated by existing migration network structures. Our results suggest that conditional transfers may be helpful in managing rural out-migration, particularly to the United States.

Insight into the determinants of both domestic and international migration and how these determinants might be mediated through public policy have been slow to emerge from the social science literature (Massey et al. 1993; Massey, Arango, et al. 1994). The issue is of central importance to Mexican and U.S. policymakers, given the historically high levels of domestic migration within Mexico and international migration of Mexicans to the United States. While this migration has brought substantial economic benefits to the U.S. and Mexican economies and to millions of Mexican families, the overall costs are high and include rapid urbanization, social dislocation, political tension, and loss of life (Diaz-Briquets 1991; Garza 1999; Kearney 1986).

In this study, we use data on Mexico's large-scale and innovative poverty-reduction program, PROGRESA, to study its impact on migration behavior. PROGRESA was designed around an experimental evaluation in which communities were randomly assigned to treatment and control groups and cash transfers were given to eligible households in the treatment communities provided that they met certain conditions. Although it was not explicitly designed to reduce rural out-migration, by altering the economic conditions of poor households at the point of origin via conditional cash transfers, PROGRESA offers a unique opportunity to evaluate the impact of changing household resources and human-capital investment on rural out-migration. The experimental design allows us to compare the difference in the behavior of treatment and control households and to be reasonably certain that the treatment itself, rather than any observable or unobservable initial conditions, leads to changes in household migration.

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Our study also complements a growing body of literature aimed at overcoming longstanding hurdles to the establishment of causal validity in migration research. Major sources of uncertainty include the researcher's limited ability to overcome issues of migrant selectivity (Chiswick 1978; Mueser 1989) and recall bias (Smith and Thomas 2003) that greatly reduce the usefulness of cross-sectional survey data on migration. This problem is exacerbated if household income is itself endogenous to prior migration of one or more household members (Stark and Taylor 1989). Recent efforts, such as the use of multilevel hazard models along with Heckman selection methods (Palloni et al. 2001) or the use of fixed-effect methods with panel data to overcome concerns about migrant selectivity (Munshi 2003), highlight important econometric methodologies that have increased our understanding of causal processes in migration research. Ideally, of course, causal validity would be best served if it were possible to use experimental data to determine the causes of migration. In this case, we are able to establish greater causal validity, although it must be recognized that our causal impact is the treatment effect of the PROGRESA antipoverty program, which includes various inputs (more on this later).

Because PROGRESA was targeted at rural poverty, the data are well suited to analyze both domestic (rural-to-urban and rural-to-rural) and international (rural–to–United States) migration. The richness of the data also enables us to examine whether PROGRESA's effect is moderated by household conditions at the start of the program, particularly household access to migrant social networks. We therefore aim to provide greater insight into two related foci of migration research. The first concerns how social relationships influence migration and the institutional context of migration decisions (Boyd 1989; Massey et al. 1993; Palloni et al. 2001). The second revolves around the varied and complex economic determinants of migration behavior (Faini and Venturini 1994; Harris and Todaro 1970; Lindstrom 1996; Stark 1991; Todaro 1969). Our intention is to evaluate the separate and complementary roles of both social and economic determinants of migration from an experimental framework using data from the conditional cash transfer program, PROGRESA.

Examining the influence of conditional transfers on rural out-migration requires careful consideration of the theoretical models of migration and the mechanism by which resource levels and transfers enter into these models. In this article, we consider three models of migration: the neoclassical model, the new economics of migration, and the network theory of migration. Each of these models is carefully considered in the next section, which focuses on how the provision of public transfers to rural households through PROGRESA is expected to alter the migration decision. Before discussing these models, however, we explain the manner in which PROGRESA operates and transfers income. We present the PROGRESA data set used for this analysis and then describe the empirical strategy used to test our hypotheses, followed by the results of that analysis. Finally, we draw conclusions and policy implications.

PROGRESA

PROGRESA was initiated in Mexico in 1997 as a mechanism for addressing extreme poverty in rural areas. A central feature of PROGRESA is the development of human capital of poor households by improving education, health, and nutrition outcomes. Two forms of cash transfers are provided to households to meet these objectives: a food grant and a school scholarship. Each component is linked to separate and independent conditional requirements. In both cases and with rare exception, transfers are provided directly to mothers under the assumption that they are more likely than fathers to use the resources to benefit their families and children.

The food grant, which is the same amount for each beneficiary household (US$16 per month as of 2001), is conditional on health check-ups for all family members and attendance at public-health lectures. Membership in the family is established at the time of a census that is conducted in a community prior to the initiation of PROGRESA and

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is used to determine household eligibility for the program.[1] At registration, households set up a schedule of health appointments for all members of the family for the year. This information is given to the clinic, and attendance records are maintained. Along with these check-ups, beneficiaries are asked to attend health and nutrition talks at the clinic. The clinic is required to fill in a form every two months. On the form, one column certifies that the household met its health check-up requirements, and one column certifies beneficiary attendance at health and nutrition talks. This form triggers the bimonthly payment to the beneficiary (Adato, Cody, and Ruel 2000). Failure of any member of the household to meet these conditions results in the loss of the food grant.[2]

School scholarships are linked to specific children and thus differ by household. The grants are awarded to mothers every two months during the school year, and all children over age 7 and under age 18 (for grades 3 through 9) are eligible. Children must register for and regularly attend school (a monthly attendance rate of 85%) to receive the grant. School officials verify registration by signing a form for each family and certify attendance by submitting attendance forms to the proper authorities. If attendance requirements are not met, the amount linked to that particular child is deducted from the bimonthly payment (Adato et al. 2000).

Because PROGRESA targets poor households, criteria were developed for determining eligibility based on household well-being. Eligible households were selected in three stages (Skoufias, Davis, and De La Vega 2001). First, potential recipient communities were identified as poor based on an index of marginality developed from the national population census. More-marginal communities were considered potential target locations and were further evaluated based on location and existence of health and school facilities. Second, a census of all households in the potential target locations was conducted to identify eligible households from within those communities. Scores were produced for each household by using discriminant analysis, and households above a certain cutoff were included as beneficiaries. Third, a list of these households was presented to the community assemblies for review and discussion, though in practice these lists were rarely modified.

The communities were randomly assigned to either treatment or control groups, and treatment began early in 1998. By the end of 1999, the year corresponding to the data in our sample, PROGRESA provided bimonthly transfers to approximately 2.3 million households, or about 40% of all rural families and 11% of all Mexican families. With the advent of the Fox administration in 2001, PROGRESA changed its name to OPORTUNIDADES and expanded operations to urban and semi-urban areas (into communities with a population of over 2,500 inhabitants). The PROGRESA/OPORTUNIDADES budget for 2002 reached US$1.9 billion, covering almost 3 million rural families and over 1.2 million urban and semi-urban families (Fox 2002; Skoufias and McClafferty 2001).

THEORIES OF MIGRATION

Public cash transfers, such as those provided by PROGRESA, may in theory influence migration decisions and modify migration patterns. The manner by which these influences operate depends both on the specifics of the program and on the mechanisms that drive migration. One condition is the requirement of physical presence, which is greatest for both children, who must attend schools and receive regular health check-ups, and the targeted beneficiary, who is normally the adult female in the household with school-aged children. Other adults in the household are required to be physically present only for annual health

1. A new census is administered after the community has been in the program for three years. This census determines, again, which households are still eligible and adds new households to the program.

2. These are the de facto rules of the program. In practice, as the program developed, PROGRESA administrators were reluctant to terminate transfers if only a few family members missed health check-ups. At the early stages of the program, which we are using in this evaluation, recipients would not have known that this was the case and likely believed that there was a probability of losing payment by missing check-ups.

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check-ups. The requirement to be physically present should affect not only labor migration but also migration by students for schooling or for marriage. Any migration of an eligible person involves a cost because it limits the ability of the household to collect transfers related to that household member. The PROGRESA conditions are not directly linked to labor supply; in fact, empirical evidence suggests that PROGRESA has no effect on labor supply among adults (Parker and Skoufias 2000). The expectation, then, is that PROGRESA will limit both labor and nonlabor migration only through the requirement of physical presence. With these factors in mind, we now consider theories of migration.

Neoclassical models of migration consider the migration decision in a cost-benefit framework in which potential migrants compare the expected utility from income at the point of origin to the expected utility from net income at possible migration destinations (Harris and Todaro 1970; Sjaastad 1962; Todaro 1969). Expectations of net income from any location depend on the characteristics of the individual, such as age, skill level, and asset position, and thus vary by individual. In a dynamic model, potential migrants should migrate if the discounted expected utility of income over some time horizon at the destination net of the costs of migrating and searching for a position exceeds the discounted expected utility of income over that same time horizon at the point of origin. If cash transfers were unconditional and there were no requirement to be physically present to receive the transfers, the receipt of transfers would not be expected to alter this calculation.

Conditionality complicates this calculation. For school-aged children, it increases the value of staying at the point of origin for the years for which they are eligible, thus reducing the probability of migration.[3] For adults, the requirement to be physically present changes the cost-benefit calculation by, in effect, increasing the cost of migration through the yearly loss of the food grant portion of PROGRESA for the household over the length of the program. This cost is incurred whether the motivation for migration is for labor or nonlabor purposes. Note that nonrecipient adults need only be physically present once per year for a health check-up. Thus, the physical-presence conditionality may have a greater effect on permanent and/or U.S. migration and a lesser effect on temporary and/or domestic migration, in which returning for check-ups is much easier. According to this version of the neoclassical model, PROGRESA is likely to reduce all types of migration for children and recipient adults but to reduce only permanent and U.S. migration for nonrecipient adults.

The discussion of the neoclassical model thus far assumes no start-up costs to migration. Suppose, alternatively, that there are monetary costs to migration and that would-be migrants are financially constrained. If the tightness of the financial constraint declines with income at a diminishing rate, the propensity to migrate as a function of income may follow an inverse-U pattern (Faini and Venturini 1993). That is, at low levels of income, additional income may relax the financial constraint, leading to greater migration; at higher levels of income, where financial constraints are less binding, additional income may reduce migration in the manner suggested in the neoclassical model. Thus, aid to a relatively poor area may increase migration by relaxing financial constraints and helping migrants cover start-up costs. Sadoulet, De Janvry, and Davis (2001) showed that in the case of PROCAMPO, another cash transfer program in rural Mexico, transfers had a dramatic impact on household liquidity. Given that PROGRESA targets rural poor households that are likely to face substantial financial constraints, the program may act to increase migration if there are substantial costs to migration.

After years of dominating the economic view of migration, the assumptions and conclusions of the neoclassical theory have been challenged by a new theory referred to as the "new economics of migration," which is based on the key insight that migration decisions

3. For school-aged children, PROGRESA may simply delay migration for the period in which they receive the transfer. However, because we are evaluating the program in its early stages, we do not expect to find this effect.

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are not necessarily made in isolation by individuals but by larger units of related people, particularly households (Massey et al. 1993). From this perspective, the decision to migrate may be considered a joint household decision, with the household sharing the costs and benefits of migration with the migrant through an explicit or implicit sharing rule. The household uses migration as one mechanism for diversifying risk and gaining access to capital in the presence of market imperfections in the credit and insurance markets (Stark and Bloom 1985; Stark and Levhari 1982). Under this model, PROGRESA's provision of cash to recipients should alter the migration decision. First, assuming the program is well managed, the cash transfer will provide a source of income that is uncorrelated with earnings in the origin sectors. As such, it improves the ability of the household to manage risk, thus reducing the need to diversify through migration. Second, as we noted earlier in the discussion of the start-up costs of migration, PROGRESA is likely to improve the household's liquidity by providing a regular (bimonthly) source of cash income.[4] According to this view of migration, PROGRESA is likely to lead to a reduction in the probability of migration for recipient households.

Based on the empirical observation that migration streams often develop in particular communities and regions, the network theory of migration highlights the importance of direct and indirect relationships in the migration decision (Boyd 1989). Migrant networks can be viewed as a migration-specific form of social capital (Massey, Goldring, and Durand 1994) that influences the migration decision in two ways. First, members of the network may provide direct assistance to migrants in the form of food, housing, transportation, or cash that reduces the cost of migration. Second, network members may provide information to potential migrants about job opportunities, safely crossing a border, and so on, which alters the idiosyncratic returns to migration. Migrant networks therefore increase the expected returns and reduce the risks and costs associated with migration.[5] As migrant networks form and thicken, they serve as catalysts for the migration of family members of network migrants as well as community members at the point of origin. Abundant evidence shows that migrant networks are positively and significantly related to migration (see, e.g., Davis, Stecklov, and Winters 2002; Davis and Winters 2001; Espinosa and Massey 1999; Massey and Espinosa 1997; Massey and Garcia España 1987; Munshi 2003; Taylor 1986; Winters, de Janvry, and Sadoulet 2001). Additional evidence indicates that the impact of networks is the greatest for U.S. migrants, for whom risks are the highest, labor market information is more scarce and costly, and the penalty for making bad forecasts is more severe (Taylor 1986). Finally, empirical studies suggest that migrant networks positively influence the economic returns to migration through higher wages and greater numbers of hours worked (Donato, Durand, and Massey 1992; Massey 1987; Neumann and Massey 1994).

Because migrant networks appear to play a significant role in migration, the effect of PROGRESA on the migration decision may depend on the presence of established networks. The theory of cumulative causation in migration (Massey, Goldring, and Durand 1994) suggests that reducing migration is increasingly difficult as migrant networks become established. Following this logic, Winters et al. (2001) found that, although lessening the marginality of a rural municipality reduces migration in general, after strong community networks are in place, this strategy is no longer effective in stemming migration flows. They argued that when migration is initiated, only a few households have network access through family networks. As migration flows increase, the community network becomes well established, substitutes for the family network, and creates a situation in which

4. The improved liquidity of the household may lead to increased migration by members of the immediate beneficiary family or may be used to help support the migration of a relative who does not need to be physically present. This could cause a shift in migration by noneligible relatives of the beneficiary household.

5. Information provided by migrant networks, such as an economic downturn or crackdown at a border, could reduce the expected returns to migration. In net, however, we assume that the information has a positive influence on returns.

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households with initially adverse characteristics for migration are able to migrate. This positive effect of networks implies that over a certain range, there is an increasing benefit of additional network migrants. Whether networks moderate the effects of PROGRESA depends largely on whether networks are nonlinear and on the size of the networks within the community at a given time. For example, in a community with strong networks, information is readily available to most individuals, reducing the risks to migration. In this case, PROGRESA transfers are more likely to relax financial constraints and allow individuals to take advantage of the information. In contrast, where networks are poorly established and risks to migration high, receiving PROGRESA will have a smaller impact because the added resources may do little to overcome the existing risks. Thus, the manner in which networks moderate the effect of PROGRESA on migration is ambiguous.

In summary, the direction of the effect of PROGRESA on the migration decision is not predictable from theory. The neoclassical model in the absence of start-up costs suggests that PROGRESA will decrease migration, but when start-up costs and financial constraints are considered, PROGRESA may help relax these constraints and induce migration. The new economics of migration theory predicts that PROGRESA will reduce migration by reducing income risk at the point of origin, limiting the need to overcome credit and insurance market failures. Given that the physical-presence requirements for adults, particularly for nonrecipient adults, are limited, the effect of PROGRESA on migration may not be strong, especially for domestic migrants, for whom returning for annual health check-ups is not difficult. This may lead to a differential effect of migration on domestic and U.S. migration. Finally, the impact of PROGRESA is hypothesized to depend on the existing structure of migration networks, particularly U.S. networks, which are considered influential in the migration decision. However, the nature of this relationship is also ambiguous.

THE PROGRESA DATA

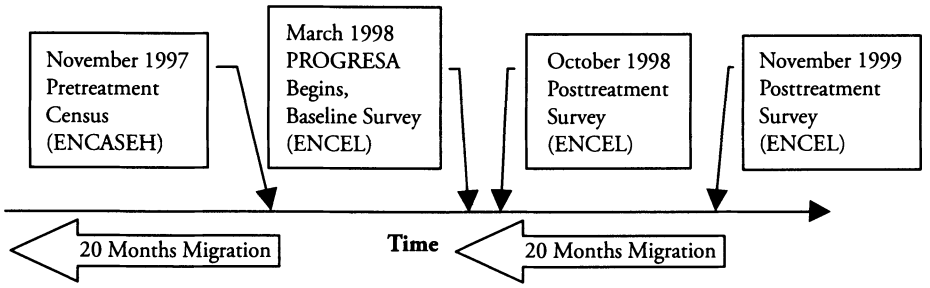
We use two primary sources of data for our empirical analysis. The first is the census (ENCASEH) conducted in November 1997 in all communities considered for participation in PROGRESA, including those later assigned to control and treatment groups. This census formed the basis for the selection of beneficiary households and included detailed household information. Because it covered all PROGRESA treatment and control communities, including households that were surveyed for the PROGRESA evaluation, the census serves as a baseline survey for this study.[6] Second, as part of an evaluation based on an experimental design, 506 PROGRESA communities in seven regions[7] were selected and randomly allocated into treatment and control groups. Only households in the treatment communities received PROGRESA for the duration of the evaluation. As part of this evaluation, follow-up surveys (ENCELs) were conducted every six months in these selected communities for approximately three years. Our analysis is primarily based on the November 1997 ENCASEH and the November 1999 ENCEL data, although we also make limited use of the October 1998 ENCEL survey. Figure 1 provides a chronology of the rounds of the collection of data included in our analysis. The ENCEL surveys collected data on all households in the 506 communities, both treatment and control, numbering over

6. A baseline household survey (ENCEL98M) was conducted in both the treatment and control communities in March 1998, prior to the initiation of PROGRESA payments in May 1998. This first ENCEL did not collect demographic, labor use, or asset information available in ENCASEH, and instead focused on household consumption. We thus use ENCASEH as the main source of data for the control variables.

7. The regional groupings are Region 1: Sierra Negra, Zongolica, and Mazateca in the states of Puebla and Veracruz; Region 2: Sierra Norte and Otomi Tepehua in the states of Hidalgo, Puebla, and Veracruz; Region 3: Sierra Gorda in the states of Hidalgo, Queretero, San Luis Potosi, and Veracruz; Region 4: Montana in the state of Guerrero; Region 5: Huasteca in the state of San Luis Potosi; Region 6: Tierra Caliente in the state of Michoacan; and Region 7: Altiplano in the state of San Luis Potosi.

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Figure 1. Timeline of PROGRESA Data Collection



24,000 total households.[8] We focus our attention on families originally classified as poor (that is, as potential PROGRESA beneficiaries).[9]

Because our interest is in how PROGRESA affects domestic and U.S. migration, we detail here our construction of the migration measures from the PROGRESA surveys. The dependent variable in our analysis refers to migration from the time of the data collection, t, and going backward to $t - n$, where n generally spans a 20-month period. Such retrospective migration data is useful because it provides data on current household members who are living elsewhere and tells us when they left the household, provided that they left within five years of the survey. However, the retrospective data on migration in PROGRESA do not provide information on repeat migrants or those who migrated during the past five years and had already returned by the time of the survey.

The data indicate that the earliest PROGRESA transfers were paid out at the end of March 1998, following the implementation of the first round of ENCEL. We use data collected in November 1999 as the source of data on posttreatment migration. In order for the November 1999 data to be an accurate reflection of migration that occurs after the initiation of PROGRESA transfers (i.e., to ensure that treatment preceded the migration), we include only migrants who left in the 20-month period preceding November 1999 (March 1998–November 1999). It is also possible to compare migration posttreatment with migration prior to PROGRESA using data from the November 1997 census (ENCASEH) that includes information about migrants who left in the 20 months before that date. The two periods of migration provide us with estimates of migration levels both before and after the initiation of PROGRESA.[10]

Descriptive statistics of the data in our analysis are shown in Table 1. The means and standard deviations are presented in the two left-hand columns for households in the treatment communities and in the middle columns for those in the control communities, and

8. Gertler (2004) noted that over the two-year experimental period, 5.5% of the households and 5.1% of the individuals dropped from the sample, but that there were no differences in attrition between the control and treatment areas, suggesting no systematic attrition bias in the analysis.

9. Initially, PROGRESA classified as eligible about 52% of households. Afterward, because of the perceived bias against certain kinds of poor households (especially elderly with no children), criteria of eligibility were revised and the program was extended to cover 78% of households. This expansion is known as *densification*. Because of the revision of the criteria for eligibility, households included in the second phase have different characteristics: these households were declared to be eligible later, and most of them started receiving cash transfers some time after the initial households. The impact of PROGRESA on their consumption could, therefore, be different. Hence, we restrict our analysis to the "predensification" poor (12,627 households).

10. A total of 1,552 households were dropped because of attrition between the 1997 and the 1999 surveys. Our results are not substantively affected if we keep those households that appear in only one of the two rounds, rather than use only households that appear in both rounds.

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Table 1. Summary Statistics for Control Variables in Treatment and Comparison Groups, by Treatment Status

	PROGRESA				
	Treatment Group		Control Group		Test of Mean Differences,
Variable	Mean	*SD*	Mean	*SD*	*p* Value
A. Control Variables					
Household size	6.01	2.43	6.03	2.41	.84
Household size squared	42.04	33.94	42.14	33.27	.93
Share of children aged 0–4	0.16	0.16	0.16	0.16	.84
Share of children aged 5–10	0.20	0.16	0.20	0.16	.80
Share of children aged 11–14	0.11	0.13	0.11	0.13	.24
Share of men aged 15–19	0.04	0.09	0.04	0.08	.26
Share of women aged 15–19	0.05	0.09	0.05	0.09	.75
Share of men aged 20–34	0.10	0.12	0.09	0.12	.15
Share of women aged 20–34	0.10	0.11	0.11	0.11	.33
Share of men aged 35–59	0.08	0.11	0.09	0.11	.27
Share of women aged 35–59	0.08	0.11	0.08	0.11	.95
Head's age	42.38	14.34	42.83	14.80	.31
Head is male	0.92	0.27	0.92	0.27	.83
Indigenous-language speaking	0.42	0.49	0.44	0.50	.70
Adult education	3.41	2.16	3.27	2.12	.30
Adult education squared	16.27	17.65	15.19	16.37	.23
Marginality index	638.82	82.76	638.45	83.66	.94
Marginality index squared	414.94	100.11	414.62	101.27	.96
Region 1	0.13	0.33	0.14	0.35	.65
Region 2	0.17	0.38	0.20	0.40	.52
Region 3	0.42	0.49	0.43	0.50	.86
Region 4	0.12	0.33	0.07	0.26	.23
Region 5	0.01	0.10	0.02	0.13	.56
Region 6	0.12	0.32	0.12	0.32	.92
Region 7	0.03	0.17	0.01	0.12	.38
B. Migration Networks					
Family domestic networks	0.02	0.18	0.01	0.15	.49
Community domestic networks	0.01	0.01	0.00	0.01	.19
Family U.S. networks	0.02	0.17	0.01	0.13	.15
Community U.S. networks	0.01	0.02	0.00	0.01	.37
C. Migration Outcomes					
Domestic migration 3/96–11/97	0.01	0.08	0.00	0.06	.28
U.S. migration 3/96–11/97	0.01	0.08	0.00	0.06	.09
Domestic migration 3/98–11/99	0.04	0.19	0.04	0.20	.82
U.S. migration 3/98–11/99	0.01	0.10	0.01	0.11	.38
Sample Size	6,745		4,170		

Notes: There were only 6,739 cases for the marginality index variables in the treatment group and 4,144 in the control group. There were 4,164 cases for the region variable in the control group.

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the number of cases is noted at the bottom. The far right columns present the p value for a test of the difference in means between treatment and control groups after we adjust for community-level clustering in the data. Because our analysis is restricted to households that are classified as poor, all households in the treatment community are eligible for treatment, and in this table and throughout the paper, the "treatment group" refers to those households that the program intended to treat.[11] Table 1 makes clear that there was very little item nonresponse in the data we use, which we handle using casewise deletion. The item nonresponse would have been much greater if we had used income and expenditure questions, which suffered much higher nonresponse levels.

Most of the variables are self-explanatory, but a few require clarification. The marginality index was calculated by PROGRESA to determine eligibility in the program. The index constitutes the PROGRESA calculation of household economic well-being before the program started; the larger the value, the wealthier the household (see Skoufias et al. 2001 for details). In creating the migrant network variables, to avoid any potential endogeneity whereby the migration of household members following the onset of PROGRESA affects network structure, we use data on migration network structures that existed before the onset of PROGRESA. Family networks are measured by person and are measured as a count of the number of household members reported to have migrated from the household in the five years prior to PROGRESA—that is, from 1992 to 1997—either within Mexico (family domestic networks) or to the United States (family U.S. networks). Estimates of community migration networks are calculated as the fraction of the labor force (adults aged 15–65) in each community, excluding the reporting household, that has migrated either within Mexico or to the United States between 1992 and 1997. An important advantage of these data is that our estimates of community migration networks are based on censuses of the selected communities, not sample surveys, so that all households are included in the community migration network variables.

The statistics from Table 1 show a few differences between treatment and control households in terms of the control variables (see Part A), but none of these differences are statistically significant at the 5% level. The mean sizes of migration networks tend to be larger for the treatment community (see Part B), but none of these differences are statistically significant.

The summary statistics for the dependent variables (Part C of Table 1), reveal a somewhat different story. Note that the migration outcome variables are the only outcomes measured at two times (in both 1997 and 1999). First, both domestic and U.S. migration levels are higher for households in the treatment group (PROGRESA) for the period March 1996–November 1997. None of these differences are statistically significant, although U.S. migration levels are higher for treatment households, and that difference is marginally significant ($p = .09$). The migration estimates for the post-PROGRESA period of March 1998–November 1999 show that the initially higher migration levels for PROGRESA households prior to treatment have evaporated, and migration levels now appear larger in control communities, although not significantly. When we estimate these differences after controlling for other differences between households, we clearly see that migration has risen over time in both treatment and control households, but migration in treatment households appears to have increased by a smaller amount, particularly for U.S. migration.[12] Our results consistently show greater estimated substantive impacts as well as higher levels of

11. The take-up ratio is approximately 96%, suggesting that the intent to treat is very similar to the actual treatment (Angelucci 2004).

12. Angelucci (2004) proposed an alternative strategy whereby the effect of PROGRESA on migration is measured using information on all labor migration in the past five years. The disadvantage of including migration five full years into the past is that it includes migration before the initiation of PROGRESA transfers. Our analyses show that migration is greater in treatment communities prior to treatment, and this likely explains Angelucci's finding that PROGRESA increases migration.

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statistical significance when we use methods that account for the initially higher levels of migration in treatment communities, as opposed to relying entirely on posttreatment migration information.

EMPIRICAL SPECIFICATION

Our theoretical model views the household decision-making process as two simultaneous decisions: one about whether to migrate domestically, and the other about whether to migrate to the United States. In theory, households can have both types of migrants, although only 13 such households appear in our data. Because the two migration decisions could be simultaneously determined, it makes sense to adopt an estimation method, such as a bivariate discrete dependent-variable model, that allows for correlated errors across the two equations. Given the small number households with both types of migrants, we test the appropriateness of such a model and conclude that the additional complexity is unnecessary.[13] The decision to migrate to the United States or within Mexico is therefore treated as two separate household decisions. Separate logistic regressions are used to estimate the effect of PROGRESA on domestic and U.S. migration. We estimate an additional model to understand better the moderating influence of migration networks on PROGRESA.

The control variables in all the specifications include a vector of individual and household variables (e.g., education, age of household head, gender of household head) that control for different prospects at the destination points in terms of potential migration, as well as any remaining differences between households prior to the onset of PROGRESA. Given the variation in income data from year to year, we use the marginality index described earlier, rather than a measure of permanent income at the point of origin.

The experimental nature of the program evaluation means that households in the treatment and control communities should be the same in terms of observable and unobservable characteristics, aside from random variation. Although our summary statistics suggest that control and treatment groups are statistically similar, earlier analysis suggests that there are significant differences between treatment and control households (Behrman and Todd 1999). We use two strategies to control further for remaining observed and unobserved factors that may affect migration behavior. First, we include a series of household characteristics in the analysis to control directly for observable differences between households. Second, we account for time-invariant unobservable differences between households in the control and treatment communities at the onset of the PROGRESA program by using a difference-in-difference (DD) estimator (Meyer 1995; Wooldridge 2002). When differences between treatment and control groups are caused by random sampling, the DD estimator has the advantage of minimizing the effects of this error as long as the effects are unrelated to the treatment and are constant over time (Wooldridge 2002).

Alongside the advantages of the DD approach, several disadvantages must also be considered (Meyer 1995). One of these disadvantages is that the DD method is traditionally estimated by using linear models. The interpretation of the DD results with nonlinear models is not straightforward because the differencing is in terms of the transformed coefficients in the model (i.e., the logit) rather than in the underlying probability. We use two methods to deal with this complication. First, we estimate the DD models by using linear probability models and compare these results to the nonlinear logit model results. The linear probability model is a problematic estimator because of heteroskedasticity and its inability to generate predictions that are bounded by 0 and 1 (Greene 2003), but it is a convenient tool to assess the consequences of nonlinearity of the logit in the DD specification.[14] Second, we

13. A convenient test compares the log-likelihood values of the bivariate probit model to the sum of the log-likelihoods of the two separate probits. In this case, the test—actually a Wald test because the models are estimated using robust standard errors—shows no advantage in estimating the bivariate probit model over a simpler model (Wald = 1.34 and $p = .25$).

14. We thank a Deputy Editor of *Demography* for suggesting this strategy.

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compare our substantive findings to those we obtain using a cross-sectional (CS) estimator on the November 1999 data, ignoring the pretreatment data. The CS estimator simply includes a dummy variable for households in treatment communities to capture the effects of PROGRESA. Although we prefer the DD specification because it accounts for variation in the levels of the outcome or explanatory variables at the onset of the experiment, the substantive results of both models are consistent.

The DD approach is also prone to error if treatment or control households are in a position to anticipate the receipt of the treatment. That is, recognizing that PROGRESA will be implemented, households alter behavior before implementation and thus the collection of baseline data. In that case, anticipation is a form of experimental contamination that may lead to biased estimation of the program effects. However, anticipation is not likely to explain our results. First, our information on migration prior to treatment is obtained from the November 1997 survey, well before the actual initiation of the program. This same survey was used to decide which households would receive transfers and which would not. Thus, until that point, households would not have known that they were eligible because the project itself had yet to declare them actual recipients. Finally, we also replicated our analysis by using pretreatment migration that was limited to the period prior to July 1996, rather than to November 1997, in order to limit any potential contaminating effects of anticipation. We found that households in PROGRESA communities showed higher levels of both domestic and U.S. migration in this period, just as they did in the 20-month period prior to the November 1997 survey, and with similar levels of significance (marginal significance in both cases for U.S. migration and nonsignificance for both cases for domestic migration). These results provide additional support that anticipation is not likely to play a major role in our estimates (results not shown but available on request).

Our main hypotheses are explored using two models: a baseline (BL) model for capturing the overall effect of PROGRESA and a migration network (MN) model that tests whether the effect of PROGRESA depends on the existing structure of family and community migration networks. Our migration outcome for household i, $M^L_{i,t}$, is dichotomous and equals 1 if any household member migrated in the 20 months prior to the survey; $L = U$ represents migration to the United States and $L = M$ means migration within Mexico, while $t = 0$ captures information about pretreatment migration captured in November 1997 and $t = 1$ captures information about posttreatment migration captured in November 1999. The above specification gives us two equations that are estimated using the logit DD model. The baseline version of this DD model, which we call DD-BL, is the following,

$$\text{Logit}(M^L_{i,t}) = b_0 + b_1 \times t + b_2 \times P_i + b_3 \times t \times P_i + c \times \mathbf{X}_i + e_{it},$$

where $M^L_{i,t} = 0$ if household i did not migrate within Mexico ($L = M$) or to the United States ($L = U$) in period t; $M^L_{i,t} = 1$ if household i migrated within Mexico or to the United States in period t; $t = 0$ for the baseline period and 1 for the posttreatment period; $P_i = 1$ if household i received PROGRESA treatment; $P_i = 0$ otherwise; $t \times P_i$ is the interaction of time and PROGRESA treatment; $\mathbf{X}_i$ is a vector of exogenous control variables; and e_{it} represents random disturbances.

In this case, there are two observations per household—one for each period—rather than one, as in the traditional CS model. The DD estimate of the effect of PROGRESA on migration is the test of the hypothesis that $b_3 = 0$ and measures the average treatment effect (Heckman, Lalonde, and Smith 1999; Skoufias 2001; Skoufias and McClafferty 2001). This estimator takes advantage of the additional information collected in the survey and can be contrasted with the basic CS estimator, which follows from dropping the terms with time, t, in DD-BL and thus allows all of the effect of PROGRESA to be captured by the coefficient on the PROGRESA treatment dummy variable.

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The MN specification focuses on determining whether the effects of PROGRESA are moderated by the existing structure of migration networks. We introduce separate terms for family and community networks in the United States and in Mexico and examine nonlinearities in the network effects. Community migration network density from the PROGRESA data indicates that the average household lives in communities where the ratio of the number of reported migrants from the past five years to the number of adult laborers is less than 1%. This relatively low average disguises considerable variation across communities: about 75% of households in communities reported no current U.S. migrants and 65% of households in communities reported no current domestic migrants, whereas about 3% of households in communities reported that 5% or more of all adult household members were in either the United States or elsewhere in Mexico. Given the nature of these data and the widespread reporting of no community networks, we use discrete variables, rather than continuous variables, to examine nonlinear effects of community networks on migration.

The importance of networks is measured by using family and community network variables that are included in the model. Interaction terms between the community network variables and the PROGRESA dummy variable are also included to test whether the effect of PROGRESA on migration is moderated by existing migration networks. In this case, the CS specification is more convenient than the DD specification because we include measures of migration networks from only a single point in time—that is, prior to treatment. Furthermore, we assume that migrant networks do not vary systematically across treatment and control households because of the random assignment. Our specification, which is separately estimated for U.S. and domestic migration, is as follows:

$$\text{Logit}(M_i) = b_0 + b_1 \times P_i + b_2 \times FN_i + b_3 \times CN_i + b_4 \times P_i \times FN_i + b_5 \times P_i \times CN_i + c \times \mathbf{X}_i + e_i .$$

Individual tests of each of the coefficients, b_2 and b_4, capture the effect of family networks (*FN*) both directly and through the interaction with PROGRESA. The coefficients b_3 and b_5 test whether community networks (*CN*) have direct effects and interaction effects on migration through PROGRESA. Both *FN* and *CN* refer to categorical variables and are thus represented by multiple dummy variables and coefficients. The **X** vector includes the remaining k covariates in the model. Significance tests on the interactions determine whether the effects of PROGRESA might be moderated by either family or community social networks. We test interactions between PROGRESA and networks by using both the linear and nonlinear specifications of the networks variables.

The network variables are introduced as categorical variables to allow for nonlinear effects without specifying particular functional relationships. Family networks are easily coded in this manner into families with no migration during the five years prior to PROGRESA (the reference category), families with one migrant (small), and families with more than one migrant (large). Few families had more than two migrants over the period of observation.

The community network variables were more difficult to code. We divided the community network variable into three separate dummy variables to capture potential nonlinear effects of community networks on migration. The smallest category, which was omitted, included communities with no network. Small-community network density was defined as less than 1%, and large was defined as greater than 1%.

RESULTS

Table 2 presents the results of the cross-sectional baseline (CS-BL) and difference-in-difference baseline (DD-BL) specifications. The top portion of each panel in the table includes the control variables (X_i) for our regression models, and the bottom portion contains the principle coefficients for tests of our hypotheses. Because the PROGRESA data were collected using two-stage cluster sampling procedures, households within the same clus-

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ter are more similar to one another than to those from other clusters (Deaton 1997). The "cluster" command from STATA was used to control for the clustering of the sample at the community level. This command calculates standard-error estimates by using a different estimator of the variance, which relaxes the assumption of independence of observations and requires only that observations be independent across clusters (STATA 2003). In all cases, reported results include this adjustment to the standard errors. In addition, Wald tests, rather than likelihood ratio tests, are used for evaluating joint significance because of the clustering in our data. In addition to providing logistic coefficient and p-value estimates, all our estimates are presented in terms of the effects of the covariates on the marginal probability of migration, where the effect is calculated at the mean value of all other control variables. Dichotomous variables are presented in terms of the effect of a shift from a value of 0 to 1 on the predicted probability.

We limit our discussion of the control variables to the coefficients that are significant and focus only on the results of the DD-BL specification. Larger household size is associated with a greater migration to both domestic and U.S. destinations, although the effect is significant only in the case of domestic migration. At the mean value of all other controls, increasing household size by about 2–3 persons leads to an increase in the probability of domestic migration by about 0.1 percentage points. The effect of household structure is also present through nine separate measures of household composition, most of which are positive and significant, although there are several negative significant coefficients as well.

Men aged 20–34 have no significant effect on migration to either the United States or within Mexico, counter to our expectation that migrants tend to be young men. Women aged 20–34, on the other hand, are associated with a strong positive increase in U.S. migration ($p = .007$). A large share of young women in the household may be a signal that men have migrated, at least to the United States.

Over 40% of household heads in our sample speak an indigenous (non-Spanish) language, a variable that proxies for cultural characteristics as well as structural poverty. The results suggest that ethnicity has opposite effects on domestic and U.S. migration. When households are indigenous-language speakers, the probability of domestic migration is not affected, but the probability of U.S. migration is lower by 0.4 percentage points ($p = .000$). These results are supported by other empirical analyses of the characteristics of Mexican migration both domestically and to the United States (Davis, Stecklov, and Winters 2002).

Education is measured as the average number of years of schooling for adults in the household. Education is a good proxy for the potential ability of households to benefit from migration through higher wage offers. Education is included as both linear and quadratic terms, and the results support this nonlinear relationship. Education appears to increase migration to both destinations significantly, while the marginal effect diminishes at higher levels of education (jointly testing both coefficients shows $p = .064$ for domestic migration, and $p = .077$ for U.S. migration). Thus, PROGRESA's long-term effect may be to increase migration through its effects on education.

The PROGRESA marginality index is included with both linear and quadratic terms. The effects of the index are insignificant in the case of domestic migration but positive for U.S. migration and jointly significant ($p = .061$). Better-off households experience greater migration to the United States, and this effect apparently increases in strength at higher wealth levels.

The regional variables are jointly significant ($p = .000$) for both domestic and U.S. migration. However, few significant differences among regions are evident when the Altiplano area of San Luis Potosi is taken as the base.

Impact of PROGRESA

We now turn our focus to the relationship between PROGRESA and migration using the DD specification (see Table 2). Three coefficients are associated with PROGRESA and

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Table 2. The Effect of PROGRESA on Domestic and U.S. Migration

	Domestic Migration			U.S. Migration		
Variables	B	$\partial P(y) / \partial x$	*p* Value	B	$\partial P(y) / \partial x$	*p* Value
A. Difference-in-Difference Specification						
Constant	−10.800	—	.000	−14.833	—	.000
Household size	0.505	0.004	.000	0.109	0.000	.342
Household size squared	−0.016	0.000	.004	0.001	0.000	.882
Share of children aged 0–4	−2.229	−0.016	.007	1.817	0.004	.178
Share of children aged 5–10	−0.293	−0.002	.687	2.365	0.005	.059
Share of children aged 11–14	0.882	0.006	.204	2.248	0.005	.080
Share of men aged 15–19	2.549	0.018	.001	3.967	0.008	.002
Share of women aged 15–19	2.641	0.019	.000	2.167	0.004	.136
Share of men aged 20–34	0.156	0.001	.839	−0.420	−0.001	.788
Share of women aged 20–34	0.350	0.003	.660	3.461	0.007	.012
Share of men aged 35–59	1.105	0.008	.096	0.204	0.000	.876
Share of women aged 35–59	3.085	0.022	.000	4.674	0.009	.000
Head's age	0.004	0.000	.532	0.032	0.000	.001
Head is male	−0.546	−0.004	.002	−0.637	−0.001	.020
Indigenous speaking	0.113	0.001	.425	−1.783	−0.004	.000
Adult education	0.211	0.002	.032	0.318	0.001	.024
Adult education squared	−0.018	0.000	.090	−0.035	0.000	.037
Marginality index	0.005	0.000	.426	0.016	0.000	.104
Marginality index squared	0.000	0.000	.506	0.000	0.000	.165
Region 1	−0.354	−0.002	.454	−1.773	−0.002	.012
Region 2	−0.057	0.000	.898	−2.330	−0.003	.003
Region 3	0.362	0.003	.400	−0.341	−0.001	.503
Region 4	−1.080	−0.005	.030	−0.500	−0.001	.422
Region 5	−0.277	−0.002	.710	0.528	0.001	.419
Region 6	−0.948	−0.005	.052	1.174	0.004	.020
Posttreatment	2.371	0.021	.000	1.274	0.003	.000
PROGRESA	0.396	0.003	.263	0.522	0.001	.117
Posttreatment × PROGRESA	−0.410	−0.003	.259	−0.864	−0.002	.034
Log pseudo-likelihood		−1,880.72			−836.14	

(continued)

migration: b_1, b_2, and b_3. The b_1 coefficients on the posttreatment variable are positive and highly significant ($p < .000$) in both the domestic and U.S. migration equations, indicating that migration levels have increased in both treatment and control communities over the two-year period from 1997 to 1999. However, the temporal rise appears to be much more substantial for domestic migration because the estimated coefficients imply an increase in the probability of domestic migration by over 2 percentage points and an increase in the probability of U.S. migration probabilities by 0.3 percentage points. The b_2 coefficient on

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(Table 2, continued)

	Domestic Migration			U.S. Migration		
Variables	B	$\partial P(y) / \partial x$	*p* Value	B	$\partial P(y) / \partial x$	*p* Value
B. Cross-Sectional Specification						
Constant	−8.724	—	.000	−14.743	—	.002
Household size	0.598	0.011	.000	0.233	0.001	.079
Household size squared	−0.019	0.000	.001	−0.002	0.000	.787
Share of children aged 0–4	−2.905	−0.054	.002	1.602	0.004	.463
Share of children aged 5–10	−0.884	−0.016	.282	2.069	0.005	.335
Share of children aged 11–14	0.787	0.015	.320	2.459	0.006	.245
Share of men aged 15–19	2.741	0.051	.001	5.216	0.013	.016
Share of women aged 15–19	3.015	0.056	.000	2.418	0.006	.305
Share of men aged 20–34	0.408	0.008	.629	2.913	0.008	.204
Share of women aged 20–34	0.755	0.014	.388	3.134	0.008	.202
Share of men aged 35–59	1.317	0.024	.085	1.314	0.003	.499
Share of women aged 35–59	2.368	0.044	.009	5.547	0.014	.016
Head's age	0.000	0.000	.956	0.025	0.000	.031
Head is male	−0.570	−0.011	.004	−0.577	−0.002	.085
Indigenous speaking	0.087	0.002	.553	−1.308	−0.003	.002
Adult education	0.218	0.004	.034	0.326	0.001	.055
Adult education squared	−0.017	0.000	.131	−0.033	0.000	.110
Marginality index	0.006	0.000	.383	0.018	0.000	.168
Marginality index squared	0.000	0.000	.432	0.000	0.000	.213
Region 1	−0.387	−0.006	.430	−2.544	−0.003	.006
Region 2	−0.230	−0.004	.620	−2.810	−0.004	.002
Region 3	0.303	0.006	.496	−0.575	−0.001	.340
Region 4	−1.077	−0.014	.039	−1.089	−0.002	.137
Region 5	−0.541	−0.008	.539	−1.137	−0.002	.249
Region 6	−1.016	−0.013	.045	1.064	0.004	.075
Posttreatment	—	—	—	—	—	—
PROGRESA	−0.024	0.000	.873	−0.391	−0.001	.096
Posttreatment × PROGRESA	—	—	—	—	—	—
Log pseudo-likelihood		−1,526.64			−511.16	

Note: $N = 10{,}883$.

the PROGRESA variable, which estimates the difference between treatment and control communities in 1997, before the onset of PROGRESA, is positive for both domestic and U.S. migration, suggesting slightly higher migration levels in PROGRESA households at the outset of the program. However, neither the domestic nor the U.S. migration coefficient is significantly different from zero, indicating that these differences are negligible and likely caused by sampling error.

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Given the DD specification, the interaction between the PROGRESA and post-treatment variables, captured by b_3, provides the statistical test of whether migration outcomes are affected by the PROGRESA program. The results in Table 2 indicate that PROGRESA has a negative effect on migration both within Mexico and to the United States. However, this coefficient is relatively small and statistically insignificant ($p = .259$) in the case of domestic migration, whereas it is much larger and significant in the case of U.S. migration ($p = .034$).

The results indicate that while migration levels were increasing over this period, PROGRESA slowed this increase for households in treatment communities. PROGRESA did little to stem the flow of rural migration to domestic, primarily urban destinations, but it did have a more substantial and significant effect on the flow of migrants to the United States. The size of the coefficient on PROGRESA in the U.S. migration models indicates that PROGRESA, after only 20 months of operation, reduced the probability of U.S. migration by about 0.2 percentage points when the other variables are set at their means. The substantive impact of the program is more obvious when we account for the fact that migration levels to the United States are quite low, particularly when spanning such a short period. In that sense, the odds of migration provide a better gauge. A simple calculation shows that the odds of U.S. migration are reduced by roughly 58% as a result of PROGRESA.

As noted earlier, the stronger results for U.S. migration are most likely driven by the conditions placed on PROGRESA recipients—namely, that nonrecipient adults, who tend to be male and generally more likely to migrate, must have a health check-up each year, or the household loses the food grant. In the case of domestic migration, this condition is relatively easy to meet. For a U.S. migrant, this condition would be more difficult and, if a migrant were unable to meet it, would impose a substantial additional cost on migration. This finding implies that cash transfers may be an important policy instrument for limiting the flow of rural migrants to international destinations. But it also suggests that conditionality that requires the physical presence of household members may substantially enhance the effectiveness of such programs, at least as long as the amount provided is large enough to affect individual and household responses.

The use of nonlinear models, such as the logit model, in the DD specification introduces a complication in that the differencing is between the underlying logits of the two models rather than between actual probabilities. One test of this approach and of the effect of the nonlinearity of our results is to replicate the model using the linear probability model. Replicating our analysis with the linear probability model provides results that are similar to our earlier DD results: PROGRESA reduces U.S. migration, although the result is now marginally significant ($p = .08$) rather than fully significant, and PROGRESA again reduces domestic migration, but not significantly (linear probability results available upon request).

Until this stage, we have treated migration as a homogenous category and have made no effort to distinguish between labor- and non-labor-related migration types. As noted earlier, because of the physical-presence requirement, PROGRESA is likely to impose a cost, and therefore to limit both types of migration. However, it is certainly possible that different migration purposes might be affected by PROGRESA in different ways. It is even possible that labor migration, for example, is positively influenced and nonlabor migration is negatively influenced by PROGRESA transfers, or vice versa, so that our failure to distinguish between the two types of migration—although accurate on the whole—blurs important distinctions between labor and nonlabor migration. However, the question of whether migration motives should be analyzed separately is more complicated than it first appears because migration motives may be difficult to determine. Results from the new economics of migration have demonstrated the economic motivations that underlie seemingly noneconomic migration patterns. Rosenzweig and Stark (1989), for example, showed that the migration of women for apparent marriage purposes in India

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plays an important role in reducing consumption variability by forging ties with faraway households. In this case, even seemingly noneconomic migration motives may be at least partly driven by underlying economic calculations. Thus, separating migration types by the stated motives may be problematic.

Nonetheless, one can disregard the above argument and test whether PROGRESA affected the stated labor and nonlabor migration motives differently.[15] Because there are so few cases of U.S. migration, further dividing the data into labor- and non-labor-related migration means that the tests are rather weak. The first test is whether PROGRESA has a different impact on labor and nonlabor migration, either domestically or to the United States. One useful test is a Wald test of whether labor- and non-labor-related migration can be combined when the domestic and international logits on migration are run. The test cannot reject the possibility that the labor and nonlabor categories should be combined in the case of domestic migration ($p = .663$). In the case of U.S. migration, the test again cannot be rejected, but the Wald statistic is closer to significance ($p = .106$). Another approach is simply to run separate analyses of the effect of PROGRESA on labor and nonlabor migration to determine whether the effect is in the same direction. In general, these tests suggest that PROGRESA effects are negative in both cases, but they are significant and larger for nonlabor migration than for labor migration. Thus, we cannot entirely discount the possibility that PROGRESA affects labor and nonlabor migration differently.

Migration Networks and Start-up Costs

The results of the network analysis are presented in Table 3, where family and community network measures are added to the basic cross-sectional regression model of Table 2 to show the effect of networks on migration. The right side of Table 3 also introduces interactions between the network measures and the PROGRESA dummy in order to determine whether the effect of PROGRESA depends on existing network structure. In order to conserve space and to facilitate the examination of the results, we exclude control variables from Table 3.

The results in Table 3 suggest that migrant networks play a strong role in the migration decision, which further substantiates results from other studies (see, e.g., Massey and Espinosa 1997; Winters et al. 2001). Domestic migration is heavily influenced by both family and community networks. Small family networks increase the probability of domestic migration by almost 2% ($p = .064$)—an estimate that is only marginally significant—whereas large family networks increase the probability by over 3% ($p = .020$). Thus, domestic family networks have a positive increasing effect on the probability of domestic migration. The effects of community networks are not linear, and the results suggest a U-shaped relationship. Small community networks reduce domestic migration by less than 1% ($p = .062$), whereas large community networks increase the probability of migration by about 1% ($p = .01$), although the former is only marginally significant.

Migration to the United States is also strongly affected by networks, but the effects are not as powerful or as significant as those for domestic migration. Small family networks barely increase migration, and this effect is not statistically significant ($p = .476$); large family networks significantly increase U.S. migration by 0.3 percentage points ($p = .015$). Community network effects on U.S. migration suggest that small network density increases migration by 0.2 percentage points ($p = .061$) and large networks increase migration by only 0.1 percentage point ($p = .095$). Although these last two coefficients are only marginally significant, they suggest that the change in U.S. migration increases with the size of U.S. community networks. However, they also offer some tentative evidence that migration effects are positively associated with the size of community networks but that the

15. The survey includes questions specifically asking whether migrants left for work, marriage, studies, or other reasons. Labor migrants are those who departed for work, and nonlabor migrants left for other reasons.

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Table 3. How Network Structures Moderate the Effect of PROGRESA on Domestic and U.S. Migration, Cross-Sectional Specifications

	Migration Network Effects						How Migration Networks Moderate PROGRESA					
	Domestic Migration			U.S. Migration			Domestic Migration			U.S. Migration		
Variables	B	∂P(*y*) / ∂*x*	*p* Value	B	∂P(*y*) / ∂*x*	*p* Value	B	∂P(*y*) / ∂*x*	*p* Value	B	∂P(*y*) / ∂*x*	*p* Value
Domestic Family Networks (small)	0.732	0.019	.064	—	—	—	–0.009	0.000	.992	—	—	—
Domestic Family Networks (large)	1.017	0.031	.020	—	—	—	0.283	0.006	.503	—	—	—
Domestic Comm. Networks (small)	–0.392	–0.006	.062	—	—	—	–0.382	–0.006	.315	—	—	—
Domestic Comm. Networks (large)	0.426	0.009	.010	—	—	—	0.554	0.012	.028	—	—	—
U.S. Family Networks (small)	—	—	—	0.402	0.001	.476	—	—	—	1.325	0.007	.176
U.S. Family Networks (large)	—	—	—	1.285	0.003	.015	—	—	—	1.564	0.009	.046
U.S. Community Networks (small)	—	—	—	0.588	0.002	.061	—	—	—	0.880	0.003	.065
U.S. Community Networks (large)	—	—	—	0.444	0.001	.095	—	—	—	0.536	0.002	.189
Domestic Family Networks (small) × PROGRESA	—	—	—	—	—	—	1.105	0.035	.251	—	—	—
Domestic Family Networks (large) × PROGRESA	—	—	—	—	—	—	0.978	0.029	.145	—	—	—
Domestic Community Networks (small) × PROGRESA	—	—	—	—	—	—	–0.006	0.000	.989	—	—	—
Domestic Community Networks (large) × PROGRESA	—	—	—	—	—	—	–0.208	–0.004	.520	—	—	—
U.S. Family Networks (small) × PROGRESA	—	—	—	—	—	—	—	—	—	–1.389	–0.002	.248
U.S. Family Networks (large) × PROGRESA	—	—	—	—	—	—	—	—	—	–0.480	–0.001	.647
U.S. Community Networks (small) × PROGRESA	—	—	—	—	—	—	—	—	—	–0.571	–0.001	.340
U.S. Community Networks (large) × PROGRESA	—	—	—	—	—	—	—	—	—	–0.136	0.000	.787
PROGRESA	–0.058	–0.001	.681	–0.368	–0.001	.105	–0.028	–0.001	.878	–0.145	0.000	.720
Log Pseudo-Likelihood		–1,510.15			–505.55			–1,508.64			–504.22	

Notes: Regressions include all the same control variables as in Table 2. $N = 10{,}883$.

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marginal effect of the networks diminishes at some point—perhaps because there is little advantage in additional information for potential migrants. In contrast, the unanticipated nonlinearity seen for domestic migration is more difficult to explain.

The results for the interactions between the network variables and PROGRESA shown in Table 3 are unambiguous in rejecting our hypothesis that social networks moderate the effect of PROGRESA. None of the interactions for either domestic or U.S. migration are even marginally significant. However, these results are not insensitive to the actual specification of networks in the model. Using a simple, linear measure of community network density in place of the categorical specification, we find both that community network density increases U.S. migration (as shown with the categorical specification for networks in Table 3) and that PROGRESA has a larger (more negative) effect—albeit only marginally significant ($p = .092$)—on U.S. migration in communities with larger U.S. community networks (not shown but available from the authors on request).

As we noted earlier, PROGRESA's effect on migration may be dependent on the wealth level of the household, for example, if there are large fixed costs to migration that very poor households are unable meet, particularly in settings such as rural Mexico, where credit markets are poorly developed. We would then expect PROGRESA to have a larger impact on migration for poorer, more-marginalized households who might otherwise find migration costs prohibitive. To examine this possibility, we estimated the models separately for different wealth categories as defined by the marginality index. Our analysis, however, showed no evidence that the effects of PROGRESA on migration varied by whether households were initially better or worse off (data not shown but available from the authors on request). This result was consistent for both domestic and U.S. migration and was not affected by the manner in which we created the wealth categories.

CONCLUSIONS

Publicly funded antipoverty programs comprising conditional cash transfers have evolved into critical instruments for fighting poverty and low levels of human capital in Latin America, including high out-migration countries such as Mexico, Honduras, and Nicaragua. The question we ask is this: Can conditional cash transfer programs reduce migration? Nowhere is this question more pertinent than in the case of Mexico. In this article, we used a large-scale experimentally designed evaluation of the PROGRESA project to examine the effect of changes in household resources and network structure on domestic and U.S. migration outcomes. Given the inherent difficulty in achieving causal inference in social research in general and in migration research in particular, this study offers an innovative approach and provides new insights into the determinants of migration.

Our results both support existing lines of understanding and raise some new questions. Increasing household income through publicly provided, conditional cash transfers reduces migration. The results are not surprising, although the magnitude of the effect is substantial. Existing research continues to expand the investigation to noneconomic motivations that underlie migration, yet it is useful to remember that economic forces matter. Even where a "culture of migration" may be deeply rooted (Kandel and Massey 2002), households appear willing to change their behaviors in the face of changing economic conditions. Conditional transfer programs apparently induce a direct effect whereby the net benefits of migration decline—at least as far as U.S. migration is concerned. Whether this means that development and income growth among rural Mexican households will also translate into lower migration levels is less clear. But our results certainly strengthen the case for more investigations into the role of economic change at the household and community level on migration patterns.

Our analysis also provides further support for the role of migration networks (Massey, Arango, et al. 1994). Networks appear to strongly influence the household's decision to migrate, at least in the case of U.S. migration. However, our analysis was not able to

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determine whether the effects of conditional transfers depend on the structure of existing migration networks. In most cases, we found no moderating effects of networks; however, in one instance when U.S. migration networks were measured using a simple, linear measure, we found that households living in communities with strong U.S. migration networks would likely decrease their U.S. migration even more following PROGRESA than those households in communities with weak networks.

The most important policy implication from our findings is that rural out-migration to international destinations may be reduced by government poverty-reduction programs like PROGRESA. This is true even in communities with strong, existing migration networks, where migration is already well established, and are thus the communities that send out the most migrants in coming years. It remains to be seen, however, whether such poverty programs can effectively reduce migration without imposing conditions, such as the requirement to enroll in and attend school or to make regular visits to a health clinic. In addition, further research should focus on whether apparent increases in educational levels among rural children brought about by PROGRESA may eventually reverse these effects by increasing migration over the long term.

REFERENCES

Adato, M., D. Cody, and M. Ruel. 2000. "An Operations Evaluation of PROGRESA From the Perspective of Beneficiaries, Promotoras, School Directors, and Health Staff." Washington, DC: International Food Policy Research Institute.

Angelucci, M. 2004. "Aid and Migration: An Analysis of the Impact of PROGRESA on the Timing and Size of Labour Migration." IZA Discussion Paper No. 1187. Institute for the Study of Labor, Bonn, Germany.

Behrman, J. and P.E. Todd. 1999. "Randomness in the Experimental Samples of PROGRESA (Education, Health, and Nutrition Program)." Washington, DC: International Food Policy Research Institute.

Boyd, M. 1989. "Family and Personal Networks in International Migration—Recent Developments and New Agendas." *International Migration Review* 23:638–70.

Chiswick, B.R. 1978. "The Effect of Americanization on the Earnings of Foreign-born Men." *Journal of Political Economy* 86:897–921.

Davis, B., G. Stecklov, and P. Winters. 2002. "Domestic and International Migration From Rural Mexico: Disaggregating the Effects of Network Structure and Composition." *Population Studies* 56:291–309.

Davis, B. and P. Winters. 2001. "Gender, Networks, and Mexico-U.S. Migration." *The Journal of Development Studies* 38(2):1–26.

Deaton, A. 1997. *The Analysis of Household Surveys*. Baltimore, MD: Johns Hopkins University Press.

Diaz-Briquets, S. 1991. "The Effects of International Migration on Latin America." Pp. 183–200 in *The Unsettled Relationship: Labor Migration and Economic Development,* edited by D.G. Papademetriou and P. Martin. New York: Greenwood.

Donato, K.M., J. Durand, and D.S. Massey. 1992. "Stemming the Tide—Assessing the Deterrent Effects of the Immigration Reform and Control Act." *Demography* 29:139–57.

Espinosa, K. and D.S. Massey. 1999. "Undocumented Migration and the Quantity and Quality of Social Capital." Pp. 106–37 in *Migration and Transnational Social Spaces,* edited by P. Ludger. Aldershot: Ashgate.

Faini, R. and A. Venturini. 1993. "Trade, Aid, and Migrations—Some Basic Policy Issues." *European Economic Review* 37(2–3):435–42.

———. 1994. "Migration and Growth: The Experience of Southern Europe." CEPR Discussion Paper No. 964. Center for Economic Policy Research, London.

Fox, V. 2002. "Informe de Gobierno." Available online at http://segundo.informe.presidencia.gob.mx/index.php?idseccion=206

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Garza, G. 1999. "Global Economy, Metropolitan Dynamics, and Urban Policies in Mexico." *Cities* 16(3):149–70.

Gertler, P.J. 2004. "Do Conditional Cash Transfers Improve Child Health? Evidence From PROGRESA's Control Randomized Experiment." *Health, Health Care and Economic Development* 94:336–41.

Greene, W.H. 2003. *Econometric Analysis.* Upper Saddle River, NJ: Prentice Hall.

Harris, J.R. and M.P. Todaro. 1970. "Migration, Unemployment, and Development—2-Sector Analysis." *American Economic Review* 60(1):126–42.

Heckman, J., R. Lalonde, and J. Smith. 1999. "Economics and Econometrics of Active Labor Market Programs." Pp. 1865–2097 in *Handbook of Labor Economics,* edited by A. Ashenfelter and D. Card. Amsterdam: Elsevier Science.

Kandel, W. and D.S. Massey. 2002. "The Culture of Mexican Migration: A Theoretical and Empirical Analysis." *Social Forces* 80:981–1004.

Kearney, M. 1986. "From the Invisible Hand to Visible Feet—Anthropological Studies of Migration and Development." *Annual Review of Anthropology* 15:331–61.

Lindstrom, D.P. 1996. "Economic Opportunity in Mexico and Return Migration From the United States." *Demography* 33:357–74.

Massey, D.S. 1987. "Do Undocumented Migrants Earn Lower Wages Than Legal Immigrants—New Evidence From Mexico." *International Migration Review* 21:236–74.

Massey, D.S., J. Arango, G. Hugo, A. Kouaouci, A. Pellegrino, and J.E. Taylor. 1993. "Theories of International Migration—A Review and Appraisal." *Population and Development Review* 19:431–66.

———. 1994. "An Evaluation of International Migration Theory: The North American Case." *Population and Development Review* 20:699–751.

Massey, D.S. and K.E. Espinosa. 1997. "What's Driving Mexico-U.S. Migration? A Theoretical, Empirical, and Policy Analysis." *American Journal of Sociology* 102:939–99.

Massey, D.S. and F. Garcia España. 1987. "The Social Process of International Migration." *Science* 237(4816):733–38.

Massey, D.S., L. Goldring, and J. Durand. 1994. "Continuities in Transnational Migration—An Analysis of 19 Mexican Communities." *American Journal of Sociology* 99:1492–533.

Meyer, B.D. 1995. "Natural and Quasi-experiments in Economics." *Journal of Business and Economic Statistics* 13(2):151–61.

Mueser, P.R. 1989. "Measuring the Impact of Locational Characteristics on Migration: Interpreting Cross-Sectional Analyses." *Demography* 26:499–513.

Munshi, K. 2003. "Networks in the Modern Economy: Mexican Migrants in the U.S. Labor Market." *Quarterly Journal of Economics* 118:549–99.

Neumann, K. and D.S. Massey. 1994. "Undocumented Migration and the Quantity and Quality of Social Capital." Presented at the annual meeting of the Population Association of American, Miami, May 5–7.

Palloni, A., D.S. Massey, M. Ceballos, K. Espinosa, and M. Spittel. 2001. "Social Capital and International Migration: A Test Using Information on Family Networks." *American Journal of Sociology* 106:1262–98.

Parker, S. and E. Skoufias. 2000. *The Impact of PROGRESA on Work, Leisure, and Time Allocation.* Washington, DC: International Food Policy Research Institute.

Rosenzweig, M.R. and O. Stark. 1989. "Consumption Smoothing, Migration, and Marriage—Evidence From Rural India." *Journal of Political Economy* 97:905–26.

Sadoulet, E., A. De Janvry, and B. Davis. 2001. "Cash Transfer Programs With Income Multipliers: PROCAMPO in Mexico." *World Development* 29:1043–56.

Sjaastad, L.A. 1962. "The Costs and Returns of Human Migration." *Journal of Political Economy* 70:80–93.

Skoufias, E. 2001. "PROGRESA and Its Impacts on the Human Capital and Welfare of Households in Rural Mexico: A Synthesis of the Results of an Evaluation by IFPRI." Research report. International Food Policy Research Institute, Washington, DC.

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Skoufias, E., B. Davis, and S. De La Vega. 2001. "Targeting the Poor in Mexico: An Evaluation of the Selection of Households Into PROGRESA." *World Development* 29:1769–84.

Skoufias, E. and B. McClafferty. 2001. "Is PROGRESA Working? Summary of the Results of an Evaluation by IFPRI." FCND Discussion Paper. International Food Policy Research Institute, Washington, DC.

Smith, J.P. and D. Thomas. 2003. "Remembrances of Things Past: Test-Retest Reliability of Retrospective Migration Histories." *Journal of the Royal Statistical Society: Series A (Statistics in Society)* 166(1):23–49.

Stark, O. 1991. *The Migration of Labor*. Cambridge: Basil Blackwell.

Stark, O. and D.E. Bloom. 1985. "The New Economics of Labor Migration." *American Economic Review* 75:173–78.

Stark, O. and D. Levhari. 1982. "On Migration and Risk in LDCs." *Economic Development and Cultural Change* 31(1):191–96.

Stark, O. and J.E. Taylor. 1989. "Relative Deprivation and International Migration." *Demography* 26:1–14.

STATA. 2003. "STATA User's Guide Release 8." College Station, TX: Stata Press.

Taylor, J.E. 1986. "Differential Migration, Networks, Information, and Risk." Pp. 147–71 in *Migration, Human Capital and Development,* edited by O. Stark. Greenwich: JAI Press.

Todaro, M.P. 1969. "Model of Labor Migration and Urban Unemployment in Less Developed Countries." *American Economic Review* 59(1):138–48.

Winters, P., A. de Janvry, and E. Sadoulet. 2001. "Family and Community Networks in Mexico-U.S. Migration." *Journal of Human Resources* 36(1):159–84.

Wooldridge, J.M. 2002. *Econometric Analysis of Cross-Section and Panel Data.* Cambridge, MA: The MIT Press.

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